visual-document-search: ColPali-style multimodal page retrieval without OCR

Akshitha Reddy Lingampally

2026-06-06

Contents

1 Abstract

We present `visual-document-search`, a minimal ColPali-style retrieval harness. Each page is encoded into per-patch visual embeddings and each query into per-token text embeddings; the score for a (query, page) pair is the MaxSim aggregation that ColBERT pioneered for text.

The package ships a deterministic synthetic encoder so the suite runs without a Vision-Language Model, with the interface matching what the real `vidore/colpali` backbone produces. We report a synthetic-corpus benchmark (25 queries × 100 pages across 5 topics): page_hit@5 = 0.24, doc_hit@10 = 0.36, MRR@10 = 0.13 — exactly the shape we'd expect with a weak (synthetic) encoder, while showing the chart-suite (topic confusion, score margin, per-topic accuracy) works end-to-end.

2 1. Background

Traditional document RAG runs OCR first, then text retrieval. This loses two things: layout information (where on the page the text was) and visual content (figures, charts, scanned tables). ColPali (Faysse et al., 2024) showed that running a Vision-Language Model directly on the page image produces per-patch embeddings that, when scored with ColBERT's MaxSim aggregation, beat OCR + text retrieval on document benchmarks like ViDoRe.

The architecture is conceptually simple: VLM encoder per page □ list of per-patch embeddings; text encoder per query □ list of per-token embeddings; score = sum over query tokens of max over patches of dot product.

3 2. Related Work

- **ColPali** (Faysse et al., 2024): the paper this project follows.
- **ColBERT** (Khattab & Zaharia, 2020): the MaxSim aggregation pattern. ColPali is the visual extension.
- **PaliGemma** (Beyer et al., 2024): the underlying VLM ColPali uses; we use a generic interface so it's swappable.

4 3. Method

4.1 3.1 Synthetic encoder

For testing without a VLM, we generate per-page patches as deterministic topic-anchored vectors plus noise:

```
anchor = hash_to_unit_vector(topic)
patches = topic_strength * anchor + (1 - topic_strength) * noise
patches /= ||patches||   # per-patch L2 normalize
```

`topic_strength = 0.6` controls signal-to-noise. The query encoder is the same shape: anchor for the query's topic + noise per query token.

4.2 3.2 MaxSim scoring

```
score(query, page) = sum over q in query_tokens of
                     max over p in page_patches of <q, p>
```

L2-normalized inputs make the dot products cosine similarities.

4.3 3.3 Metrics

- `page_hit@k`: 1 if any of the top-k pages is a relevant page (the strictest metric)
- `doc_hit@k`: 1 if any of the top-k pages is from a relevant document (more forgiving when a doc has multiple pages)
- `recall@k`: fraction of relevant pages in top-k
- `mrr@k`: 1 / rank of first relevant page

5 4. Data

In-repo synthetic corpus: 5 topics × 5 documents per topic × 4 pages per doc = 100 pages. 25 queries, one per (topic, doc). Each query targets a specific (topic, doc) pair; the 4 pages of that doc are the relevant set.

Real-data mode: swap the synthetic encoder for `vidore/colpali` and plug into a ViDoRe-style PDF benchmark.

6 5. Evaluation Setup

Standard run: 25 queries × top-10 retrieval over 100 pages. Hardware: Apple M-series CPU.

7 6. Results

metric	value
page_hit@1	0.080
page_hit@5	0.240
page_hit@10	0.360
doc_hit@5	0.240
recall@10	0.150
mrr@10	0.131

The synthetic encoder is intentionally weak (topic_strength = 0.6 plus heavy noise), so the chart *shapes* are interesting, not the absolute numbers. Real ColPali on a real PDF benchmark lands page_hit@5 around 0.70-0.85 on ViDoRe (per Faysse et al., 2024); that gap is the value of the trained backbone over the topic-anchored synthetic stand-in.

8 7. Ablations

`topic_strength` sweep □ {0.4, 0.6, 0.8}: page_hit@5 lifts from 0.08 (very noisy) to 0.60 (clean topic signal); a sanity check that the retrieval pipeline reflects encoder quality.

9 8. Discussion

The harness's value is its end-to-end shape, not the synthetic numbers. Once a real VLM is plugged in, the same five charts surface the things that matter for production document retrieval: per-topic accuracy (the "which subject matter the model is good at"), score margin (the "how confident the rank-1 was"), topic confusion (the "which subject matter gets confused with which"). All five run on the same per-query JSONL artifact, so swapping encoders re-uses the chart code unchanged.

10 9. Limitations

1. **Synthetic encoder.** Real ColPali needs `colpali-engine` + a GPU. The harness is set up to drop it in but we don't ship the integration.
2. **No PDF □ page-image pipeline.** Production needs `pdf2image` + PIL.
3. **Brute-force MaxSim.** Scales O(corpus × patches × q_tokens); above ~10K pages use ColBERTv2-style PLAID indexing.
4. **Page-level retrieval only.** Doc-level aggregation is "any page = doc hit"; a weighted-sum-per-doc variant would lift doc_hit on long-doc corpora.

11 10. Future Work

- ☐ Real ColPali backbone (`vidore/colpali`) + ViDoRe benchmark.
- ☐ PLAID indexing (centroids + IVF) for scale.
- ☐ Multi-page query (question spans multiple pages).
- ☐ Compare against OCR + BM25 baseline (the ColPali headline gap).

12 11. References

- Beyer, L., et al. (2024). *PaliGemma: A versatile 3B VLM for transfer.* arXiv:2407.07726.
- Faysse, M., et al. (2024). *ColPali: Efficient Document Retrieval with Vision Language Models.* arXiv:2407.01449.
- Khattab, O., & Zaharia, M. (2020). *ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT.* SIGIR.

13 Appendix A. Reproducibility

- Repo: `Akshitha024/visual-document-search`, MIT.
- Reproduce: `make bench && make plots`.
- 5 charts in `results/figures/`.
- Test artifacts in `docs/test_results/`.